INTERNATIONAL SEARCH REPORT

International application No.
PCT/US 22/13213

A. CLASSIFICATION OF SUBJECT MATTER

IPC - G01N 33/50, G01N 33/68, C12N 15/85 (2022.01)

CPC - G01N 33/5011, C12Q 1/6806, G01N 33/6875, C07K 23/13

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
See Search History document

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched
See Search History document

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
See Search History document

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2020/232416 A1 (WHITEHEAD INSTITUTE FOR BIOMEDICAL RESEARCH) 19 November 2020 (19.11.2020) Claim 1, para [0004], [0058], [0059], [0060], [0066], [0143], [0188], [0190], [0193], [0198], [0436]	1-4, 30, 58-63
A	SONG et al., Client proximity enhancement inside cellular membrane-less compartments governed by client-compartment interactions. Nature Communication, November 2020, Vol 11, No 1, pp: 1-13. whole doc.	1-4, 30, 58-63

☐ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
17 May 2022	JUN 28 2022
Name and mailing address of the ISA/US Mail Stop PCT, Attn: ISA/US, Commissioner for Patents P.O. Box 1450, Alexandria, Virginia 22313-1450 Facsimile No. 571-273-8300	Authorized officer Kari Rodriquez Telephone No. PCT Helpdesk: 571-272-4300

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Box No. I Nucleotide and/or amino acid sequence(s) (Continuation of item 1.c of the first sheet)

1. With regard to any nucleotide and/or amino acid sequence disclosed in the international application, the international search was carried out on the basis of a sequence listing:

   a. [X] forming part of the international application as filed:

      [X] in the form of an Annex C/ST.25 text file.

      [ ] on paper or in the form of an image file.

   b. [ ] furnished together with the international application under PCT Rule 13*ter*.1(a) for the purposes of international search only in the form of an Annex C/ST.25 text file.

   c. [ ] furnished subsequent to the international filing date for the purposes of international search only:

      [ ] in the form of an Annex C/ST.25 text file (Rule 13*ter*.1(a)).

      [ ] on paper or in the form of an image file (Rule 13*ter*.1(b) and Administrative Instructions, Section 713).

2. [ ] In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that forming part of the application as filed or does not go beyond the application as filed, as appropriate, were furnished.

3. Additional comments:

INTERNATIONAL SEARCH REPORT

International application No.
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Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☒ Claims Nos.: 5-29, 37-57, 64-91, 95-101
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

- see extra sheet for Box No. III Observations where unity of invention is lacking -

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☒ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:
1-4, 30, 58-63 limited to a target peptide fused to a fluorophore

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☐ No protest accompanied the payment of additional search fees.

Continuation of:
Box No. III. Observations where unity of invention is lacking

This application contains the following inventions or groups of inventions which are not so linked as to form a single general inventive concept under PCT Rule 13.1. In order for all inventions to be searched, the appropriate additional search fees must be paid.

Groups I+: Claims 1-4, 30-36, 58-63, 92-94, drawn to a method of identifying an inhibitory activity of an exogenous agent that disrupts binding of a target peptide to a biomolecule. The method will be searched to the extent that the target molecule encompasses a target peptide fused to a fluorophore (see Applicant's Specification, para [0039] and Fig. 1A.) It is believed that claims 1-4, 30, 58-63 encompass this first named invention, and thus these claims will be searched without fee to the extent that they encompass a target peptide fused to a fluorophore. Additional target molecule(s) will be searched upon the payment of additional fees. Applicants must specify the claims that encompass any additionally elected target molecule(s). Applicants must further indicate, if applicable, the claims which encompass the first named invention, if different than what was indicated above for this group. Failure to clearly identify how any paid additional invention fees are to be applied to the "+" group(s) will result in only the first claimed invention to be searched. An exemplary election would be a target molecule comprising a target peptide and a light-sensitive receptor and/or fluorophore (see Applicant's Specification, para [0039] and Fig. 1C.) (Claims 1-4, 31, 34/(in part), 58-63, 92).

Another exemplary election would be a target molecule comprising a target peptide, a light-sensitive receptor and a cognate partner of said peptide fused to a IDR and/or fluorophore (see Applicant's Specification, para [0039] and Fig. 1D.) (Claims 1-4, 32, 34/(in part), 35, 58-63, 93).
Another exemplary election would be a target molecule comprising a target peptide, a cognate partner of a light-sensitive receptor, and a light-sensitive receptor fused to a IDR and/or fluorophore (see Applicant's Specification, para [0039] and Fig. 1E.) (Claims 1-4, 33, 34/(in part), 36, 58-63, 94).

The inventions listed as Groups I+ do not relate to a single general inventive concept under PCT Rule 13.1 because, under PCT Rule 13.2, they lack the same or corresponding special technical features for the following reasons:

Special Technical Features

The inventions of Groups I+ each include the special technical feature of a specific target molecule comprising a target peptide fused to selected peptide or fluorophore, not required by any of the other inventions of Groups I+.

Common Technical Features

The inventions of Groups I+ share the technical feature of a method of identifying an inhibitory activity of an exogenous agent that disrupts binding of a target peptide to a biomolecule.

However, these shared technical features do not represent a contribution over prior art in view of WO 2020/232416 A1 to Whitehead Institute for Biomedical Research (hereinafter "Whitehead Inst").

Whitehead Inst teaches (instant claim 1) a method of identifying an inhibitory activity of an exogenous agent that disrupts binding of a target peptide to a biomolecule (Claim 1, A method of characterizing an agent, comprising contacting the agent with a composition comprising a condensate having at least one component, and measuring incorporation of the agent in the condensate..; para [0198], a high throughput screen (HTS) is performed to characterize a plurality of agents and/or a plurality of different condensates.), the method comprising:
(i) contacting a cell with said agent, wherein said cell comprises a target molecule that comprises said target peptide (para [0058], FIG. 24 illustrates ER/MED1 droplets contacted with a fluorescent peptide (left side). Upon estrogen exposure, the Estrogen Receptor undergoes a conformational change that allows it to interact with MED1 FXXFF domain (right side).);
(ii) detecting membrane-less bodies comprising said target molecule, wherein said target molecule is incorporated into said membrane-less bodies when said target peptide is released from said binding to said biomolecule and undergoes phase separation at a subregion of said cell that is separate and distinct from a sub-region where said target peptide binds to said biomolecule (para [0059], FIG. 25 shows that upon addition of the FXXFF peptide (QNPIFTSFFQITG; SEQ ID NO: 1) to ER/MED1 droplets, the peptide is incorporated into MED1 droplets resulting in a decrease in partitioning of ER into the MED1 droplet.; Note, FXXFF peptide, MED1 and ER correspond to inhibitor, target peptide and biomolecule, respectively, of instant claim.);
wherein said detecting comprises measuring a change in an intensity and a size of said membrane-less bodies (para [0188], methods of assessing whether differential expression of one or more condensate components by a cell resistant to an agent causes or contributes to the resistance.; [0193], In some embodiments, the method comprises determining whether the second agent counters the effects of the resistance to the agent (e.g., drug resistance) caused by the first agent (e.g., it is determined whether contact with the second agent reduces the size of or eliminates the condensate; Note, reduce the size of condensate leads to change the intensity of said condensate.).

Although Whitehead Inst does not specifically teach wherein a bigger change in said intensity and said size of said membrane-less bodies relative to a change in an intensity and a size of membrane-less bodies in the absence of said agent indicates that said agent disrupts binding of said target peptide to said biomolecule. However, it would have been obvious to one of ordinary skill in the art to have correlated the changes in the size and intensity of membrane-less bodies with the ability of an inhibitory agent in disrupting the binding between the target molecule and biomolecule.

----continued on next sheet----

Continuation of:
Box No. III. Observations where unity of invention is lacking

Some inventions of Groups I+ share the technical feature of a target peptide fused to a fluorophore which is disclosed by Whitehead Inst (para [0048], FIG. 14B shows the results of a Fluorescence Recovery Aller Photobleaching (FRAP) assay with fluorescently labeled BRD4.

Some inventions of Groups I+ share the technical feature of a target peptide fused to a IDR which is disclosed by Whitehead Inst (para [0134], the component of the condensate is an RNA polymerase associated with mRNA initiation and elongation.....In some embodiments, the RNA polymerase II C-terminal region comprises an intrinsically disordered region (IDR).).

Some inventions of Groups I+ share the technical feature of light-sensitive receptor, which is disclosed by WO 2019/147611 A2 to The Trustee of Princeton University (hereinafter "Princeton Univ") (Claim 1, 1. A system for the controlled clustering of gene regulatory proteins at specific target genomic loci, comprising: a first structure comprising a Cas-based genomic targeting protein fused or attached to one or more first sequences, each first sequence including at least one sequence selected from the group consisting of a light-sensitive receptor.).

As said technical features were known in the art at the time of the invention, these cannot be considered special technical features that would otherwise unify the groups.

Groups I+ therefore lack unity under PCT Rule 13 because they do not share a same or corresponding special technical feature.

Note, Claim 36 which depend from "the method of claim 3333", is reconstrued as though depending from "the method of claim 33 ".

Item 4 (continued):
Claims 5-29, 37-57, 64-91, 95-101 are held unsearchable because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).